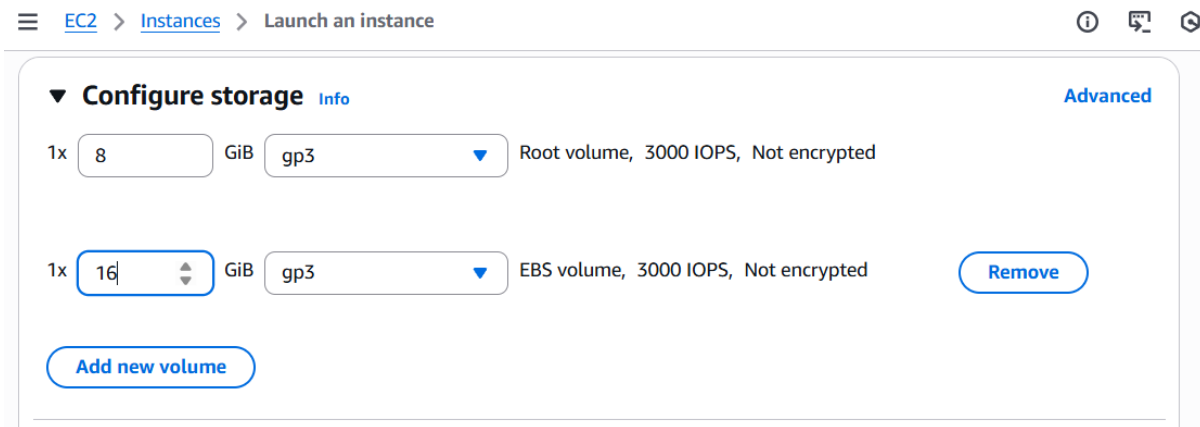


AWS Cloud Practical

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1. EC2 and EBS Practical

- Create an EC2 with extra 16 GiB EBS storage.



- Login to EC2 using MobaXterm/PuTTY.

```
[ec2-user@ip-172-31-12-249 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0   8G  0 disk 
├─nvme0n1p1 259:2    0   8G  0 part /
├─nvme0n1p127 259:3    0   1M  0 part 
├─nvme0n1p128 259:4    0  10M  0 part /boot/efi
└─nvme1n1   259:1    0  16G  0 disk 
[ec2-user@ip-172-31-12-249 ~]$ sudo mkfs -t ext4 /dev/nvme1n1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 4194304 4k blocks and 1048576 inodes
Filesystem UUID: 05fe6e19-1258-4e5a-bc24-70f7e135a5b9
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-172-31-12-249 ~]$ mkdir -p /home/ec2-user/test
[ec2-user@ip-172-31-12-249 ~]$ sudo mount /dev/nvme1n1 /home/ec2-user/test
[ec2-user@ip-172-31-12-249 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0   4.0M   0% /dev
tmpfs           459M   0  459M   0% /dev/shm
tmpfs           184M 444K  183M   1% /run
/dev/nvme0n1p1  8.0G  1.6G  6.5G  20% /
tmpfs           459M   0  459M   0% /tmp
/dev/nvme0n1p128 10M   1.3M   8.7M  13% /boot/efi
tmpfs           92M   0   92M   0% /run/user/1000
/dev/nvme1n1    16G   24K   15G   1% /home/ec2-user/test
[ec2-user@ip-172-31-12-249 ~]$ cd /home/ec2-user/test
```

Check attached volumes

Additionally attached volume

Format the new volume

Create mount directory

Mount the volume

```
[ec2-user@ip-172-31-12-249 test]$ sudo chown -R ec2-user:ec2-user /home/ec2-user/test
[ec2-user@ip-172-31-12-249 test]$ cd /home/ec2-user/test
[ec2-user@ip-172-31-12-249 test]$ nano file1.txt
[ec2-user@ip-172-31-12-249 test]$ ls
file1.txt  lost+found
[ec2-user@ip-172-31-12-249 test]$ cat file1.txt
This is EBS volume test data
```

Create data

```
[ec2-user@ip-172-31-12-249 test]$ cd ..
[ec2-user@ip-172-31-12-249 ~]$ sudo umount /home/ec2-user/test
[ec2-user@ip-172-31-12-249 ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1             259:0    0   8G  0 disk
├─nvme0n1p1         259:2    0   8G  0 part /
├─nvme0n1p127       259:3    0   1M  0 part
├─nvme0n1p128       259:4    0  10M  0 part /boot/efi
└─nvme1n1           259:1    0  16G  0 disk
[ec2-user@ip-172-31-12-249 ~]$
```

Unmount the volume

- Data stays on the volume, mount is removed.
- Then detach the volume from the first EC2.
 - AWS Console → EC2 → Volumes

Volumes (1/4) Info Last updated 2 minutes ago Recycle Bin Actions Create volume

Saved filter sets Choose filter set Search

	Name	Volume ID	Type	Size
☐		vol-0ea75080a01cff912	gp3	8 GiB
☐		vol-0bd8579a0d4925ee6	gp3	8 GiB
☒		vol-02882554fe2558385	gp3	16 GiB
☐		vol-02ba1553fa2aff8b0	gp3	16 GiB

Volume ID: vol-02882554fe2558385

Details | Status checks | Monitoring | Tags

Volume ID	Size	Type	Status check

Modify volume
Create snapshot
Create snapshot lifecycle policy
Delete volume
Attach volume
Detach volume
Force detach volume
Manage auto-enabled I/O
Copy volume - new
Manage tags
Resilience testing
Previously fault injection

- Launch second EC2 in the same region as first EC2.

EC2 > Instances > Launch an instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Recents | **Quick Start**

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

- Then attach the 16 GiB volume to the second EC2 which is detached from the first EC2.

Volumes (1/5) [Info](#) Last updated less than a minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets: Choose filter set [Search](#)

	Name	Volume ID	Type	Size
☐		vol-0ea75080a01cff912	gp3	8 GiB
☐		vol-0bd8579a0d4925ee6	gp3	8 GiB
☒		vol-02882554fe2558385	gp3	16 GiB
☐		vol-09e36a9ac89984cfe	gp3	8 GiB
☐		vol-02ba1553fa2aff8b0	gp3	16 GiB

Modify volume

Create snapshot

Create snapshot lifecycle policy

Delete volume

Attach volume

Detach volume

Force detach volume

Manage auto-enabled I/O

Copy volume - new

Manage tags

Resilience testing

Previously fault injection

Volume ID: vol-02882554fe2558385

Snapshot ID
snap-0c2ca01
snap-0daaa62
-
snap-0daaa62
-

EC2 > Volumes > vol-02882554fe2558385 > Attach volume

Attach volume Info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
vol-02882554fe2558385

Availability Zone
eu-n1-az1 (eu-north-1a)

Instance Info
i-08f8269fdd8b3e144 (EC2-EBS-Demo-2) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name Info
/dev/xvdf

Recommended device names for Linux: /dev/xvda for root volume. /dev/sd[f-p] for data volumes.

ⓘ Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

Cancel Attach volume

- Login to second EC2 instance and mount the volume to verify the data

```

16.170.155.201 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...
/home/ec2-user/second_instance
Name
lost+found
file1.txt

MobaXterm Personal Edition v25.4
(SSSH client, X server and network tools)

> SSH session to ec2-user@16.170.155.201
  Direct SSH      : ✓
  SSH compression : ✓
  SSH-browser     : ✓
  X11-forwarding  : ✗ (disabled or not supported by server)

> For more info, ctrl+click on help or visit our website.

#####
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-14-59 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1      259:0    0   8G  0 disk
├─nvme0n1p1  259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1      259:4    0  16G  0 disk

[ec2-user@ip-172-31-14-59 ~]$ mkdir -p /home/ec2-user/second_instance
[ec2-user@ip-172-31-14-59 ~]$ sudo mount /dev/nvme1n1 /home/ec2-user/second_instance
[ec2-user@ip-172-31-14-59 ~]$ cd /home/ec2-user/second_instance
[ec2-user@ip-172-31-14-59 second_instance]$ ls
file1.txt  lost+found
[ec2-user@ip-172-31-14-59 second_instance]$ cat file1.txt
This is EBS volume test data
[ec2-user@ip-172-31-14-59 second_instance]$

```

- An EBS volume can be detached from one EC2 instance and attached to another instance in the same Availability Zone while preserving data.